Data is all around you and has occupied such a great role in your life that you probably don't even notice it most of the times. The madness of machine learning is one small drop in the ocean, making some of those roles possible. Everything from a smart chatbot to a recommendation system on your favourite online store runs at the helm of some smart algorithms designed to make the user experience seamless and simpler.

Getting straight to the basics

Picture the typical case of a user, just like you, waddling through websites, generating content, sharing opinions and creating enormous quantities of data. This data may just seem like a sequence of strings and numbers but when applied to a computing system can yield some magnificent results.

In the case of machine learning, a team of engineers and analysts first feed the data to softwares like SPSS and MATLAB. The objective here becomes to feed both the inputs and desired outputs to the system so that the machine can traverse through the building blocks and ultimately 'learn' how to arrive at the output.

From a technical standpoint, this is done through a system of layers and nodes that shoot up the informational input with different weights. These nodes can represent data points and can affect the accuracy of the final model. The primary objective of machine learning thus becomes to test the models by tweaking the parameters and comparing the produced results to the actual results.

Time to brush up some statistical analysis. Remember all those classes spent learning about standard deviation in school. The standard deviation becomes a tool to assess how close the values of the model match that of the actual values.

You might be wondering how this is done in such a manner? Analysts split up portions of the data with one part for 'training' the data and one for 'testing it. In this context, training and testing allude to letting the system learn the trained data or train it, then juxtaposing it against the testing data for accuracy.

Components of a typical network

You'll be hearing the words 'neural networks' being repeated many times throughout any machine learning text. It's because a machine learning algorithmic system closely matches that of an organic neural network where